

Section 1: Introduction to the Cell

1. **The basic structural and functional unit of all living organisms is the:**
 - A) Tissue
 - **B) Cell**
 - C) Organ
 - D) Atom
2. **Who was the first scientist to observe "cells" in a piece of cork?**
 - **A) Robert Hooke**
 - B) Antonie van Leeuwenhoek
 - C) Robert Brown
 - D) Rudolf Virchow
3. **The Cell Theory was primarily proposed by:**
 - A) Watson and Crick
 - **B) Schleiden and Schwann**
 - C) Darwin and Lamarck
 - D) Mendel and Morgan
4. **According to Cell Theory, all cells arise from:**
 - A) Non-living matter
 - B) Spontaneous generation
 - **C) Pre-existing cells**
 - D) Only air and water
5. **Which instrument is essential for studying the internal structure of a cell?**
 - A) Telescope
 - B) Stethoscope
 - **C) Microscope**
 - D) Periscope
6. **Organisms consisting of only one cell are called:**
 - **A) Unicellular**
 - B) Multicellular
 - C) Acellular
 - D) B cellular
7. **Cells that lack a defined nucleus and membrane-bound organelles are:**

- **A) Prokaryotic**
 - B) Eukaryotic
 - C) Animal cells
 - D) Plant cells
8. **Bacteria are an example of:**
- **A) Prokaryotic cells**
 - B) Eukaryotic cells
 - C) Viruses
 - D) Fungi
9. **The study of cells is known as:**
- A) Histology
 - **B) Cytology (Cell Biology)**
 - C) Physiology
 - D) Anatomy
10. **The "living" substance within the cell membrane is called:**
- A) Cytoskeleton
 - **B) Protoplasm**
 - C) Nucleoplasm
 - D) Ectoplasm

Section 2: Cell Structures and Organelles

11. **The outer boundary of an animal cell is the:**
- **A) Cell Membrane**
 - B) Cell Wall
 - C) Nuclear Envelope
 - D) Cytoplasm
12. **The rigid outer layer found only in plant cells is the:**
- A) Cell Membrane
 - **B) Cell Wall**
 - C) Vacuole
 - D) Chloroplast
13. **The Jelly-like substance between the nucleus and cell membrane is:**
- **A) Cytoplasm**
 - B) Mitochondria
 - C) Ribosome
 - D) Lysosome

14. **Which organelle is known as the "Powerhouse of the cell"?**
- A) Nucleus
 - B) Golgi Body
 - **C) Mitochondrion**
 - D) Ribosome
15. **The "Control Center" of the cell containing genetic material is the:**
- **A) Nucleus**
 - B) Nucleolus
 - C) Centriole
 - D) Vacuole
16. **Green-colored organelles found in plants for photosynthesis are:**
- A) Leucoplasts
 - B) Chromoplasts
 - **C) Chloroplasts**
 - D) Amyloplasts
17. **Protein synthesis occurs in which organelle?**
- A) Lysosome
 - **B) Ribosome**
 - C) Smooth ER
 - D) Vacuole
18. **The network of channels involved in the transport of materials is the:**
- **A) Endoplasmic Reticulum (ER)**
 - B) Golgi Apparatus
 - C) Mitochondria
 - D) Nucleus
19. **Organelles containing digestive enzymes, often called "Suicide Bags":**
- **A) Lysosomes**
 - B) Peroxisomes
 - C) Ribosomes
 - D) Centrioles
20. **Large central vacuoles are a characteristic feature of:**
- **A) Plant cells**
 - B) Animal cells

- C) Bacterial cells
- D) Human blood cells

Section 3: Cell Division and Function

21. The process by which a cell divides into two identical daughter cells is:
- **A) Mitosis**
 - B) Meiosis
 - C) Fertilization
 - D) Osmosis
22. Meiosis results in the formation of:
- A) Somatic cells
 - **B) Gametes (Sperm/Egg cells)**
 - C) Nerve cells
 - D) Bone cells
23. The number of chromosomes in a human somatic cell is:
- A) 23
 - **B) 46**
 - C) 44
 - D) 48
24. Chromosomes are made up of:
- **A) DNA and Proteins**
 - B) RNA only
 - C) Fats and Carbohydrates
 - D) Enzymes
25. The thread-like structures seen in the nucleus during cell division are:
- A) Ribosomes
 - **B) Chromosomes**
 - C) Mitochondria
 - D) Nucleoli

Section 4: Advanced Concepts & Organelle Roles

26. Which organelle packages and secretes chemical substances?
- A) Mitochondria
 - **B) Golgi Apparatus**

- C) Ribosome
 - D) Vacuole
27. **Rough Endoplasmic Reticulum (RER) is called "rough" because of:**
- A) Its shape
 - **B) Attached Ribosomes**
 - C) Attached Mitochondria
 - D) Thick walls
28. **The movement of water from high concentration to low concentration through a membrane is:**
- A) Diffusion
 - **B) Osmosis**
 - C) Active Transport
 - D) Circulation
29. **ATP is generated in which organelle?**
- **A) Mitochondria**
 - B) Nucleus
 - C) Chloroplast
 - D) Ribosome
30. **Which structure is found in animal cells but usually not in plant cells?**
- **A) Centrioles**
 - B) Cell wall
 - C) Large vacuole
 - D) Chloroplast

Section 5: Rapid Fire Review (31-100)

- 31. **Unit of inheritance? (Ans: Gene)**
- 32. **DNA stands for? (Ans: Deoxyribonucleic Acid)**
- 33. **Main component of plant cell wall? (Ans: Cellulose)**
- 34. **Does a red blood cell have a nucleus when mature? (Ans: No)**
- 35. **Fluid mosaic model describes what? (Ans: Cell Membrane)**
- 36. **Division of the nucleus is called? (Ans: Karyokinesis)**
- 37. **Division of the cytoplasm is called? (Ans: Cytokinesis)**
- 38. **Smallest cell in the human body? (Ans: Sperm cell)**
- 39. **Largest cell in the human body? (Ans: Egg cell/Ovum)**

40. **Longest cell in the human body?** (Ans: **Nerve cell/Neuron**)
41. **Who discovered the nucleus?** (Ans: **Robert Brown**)
42. **Total stages of Mitosis?** (Ans: **4 - Prophase, Metaphase, Anaphase, Telophase**)
43. **During which phase do chromosomes align in the center?**
(Ans: **Metaphase**)
44. **Sister chromatids separate during?** (Ans: **Anaphase**)
45. **What is a "diploid" cell?** (Ans: **Cell with full set of chromosomes**)
46. **What is a "haploid" cell?** (Ans: **Cell with half set of chromosomes**)
47. **Number of chromosomes in human sperm?** (Ans: **23**)
48. **Organelle responsible for lipid synthesis?** (Ans: **Smooth ER**)
49. **What protects the cell from the outside environment?** (Ans: **Cell Membrane**)
50. **Do plant cells have mitochondria?** (Ans: **Yes**)
51. **Main function of a vacuole?** (Ans: **Storage of waste/water**)
52. **What is the "Nucleolus" involved in?** (Ans: **Ribosome production**)
53. **Pores in the nuclear membrane?** (Ans: **Nuclear pores**)
54. **Substance that makes up the cell wall of fungi?** (Ans: **Chitin**)
55. **Double membrane organelles?** (Ans: **Nucleus, Mitochondria, Chloroplasts**)
56. **Single membrane organelle?** (Ans: **Lysosome/Vacuole**)
57. **Organelle without a membrane?** (Ans: **Ribosome**)
58. **Function of Plastids?** (Ans: **Storage/Photosynthesis**)
59. **Colorless plastids are?** (Ans: **Leucoplasts**)
60. **Colored plastids (other than green) are?** (Ans: **Chromoplasts**)
61. **Semi-permeable membrane allows what?** (Ans: **Only certain substances to pass**)
62. **Example of a eukaryotic cell?** (Ans: **Onion cell/Human cheek cell**)
63. **Does osmosis require energy?** (Ans: **No**)
64. **Active transport requires?** (Ans: **Energy/ATP**)

65. **Physical basis of life?** (Ans: **Protoplasm**)
66. **Chromosome count in a Down syndrome patient?** (Ans: **47**)
67. **Chromatin is found in?** (Ans: **The Nucleus**)
68. **What is 'Turgor Pressure'?** (Ans: **Pressure of vacuole against cell wall**)
69. **Are viruses considered true cells?** (Ans: **No**)
70. **Shortest phase of mitosis?** (Ans: **Anaphase**)
71. **What is "Crossing over"?** (Ans: **Exchange of genetic material in meiosis**)
72. **What creates the spindle fibers?** (Ans: **Centrioles**)
73. **Is the cell wall living?** (Ans: **No, it is non-living**)
74. **Main sugar used by cells for energy?** (Ans: **Glucose**)
75. **What is ATP?** (Ans: **Energy currency of the cell**)
76. **Where is DNA stored in bacteria?** (Ans: **Nucleoid region**)
77. **Which organelle detoxifies poison?** (Ans: **Smooth ER**)
78. **What is 'Diffusion'?** (Ans: **Movement of particles from high to low concentration**)
79. **Primary pigment in chloroplasts?** (Ans: **Chlorophyll**)
80. **Shape of plant cells?** (Ans: **Usually rectangular/Fixed**)
81. **Shape of animal cells?** (Ans: **Irregular/Round**)
82. **Does an Amoeba have a nucleus?** (Ans: **Yes**)
83. **Smallest living cell?** (Ans: **Mycoplasma**)
84. **Largest single cell in the world?** (Ans: **Ostrich Egg**)
85. **Function of Cristae?** (Ans: **Increase surface area in mitochondria**)
86. **Who said "omnis cellula e cellula"?** (Ans: **Rudolf Virchow**)
87. **What happens to a cell in distilled water?** (Ans: **It swells/bursts**)
88. **What happens to a cell in salt water?** (Ans: **It shrinks**)
89. **What is 'Phagocytosis'?** (Ans: **Cell eating**)
90. **What is 'Pinocytosis'?** (Ans: **Cell drinking**)
91. **Do animal cells have plastids?** (Ans: **No**)
92. **Main function of lysosomes?** (Ans: **Intracellular digestion**)
93. **What are "Genes" made of?** (Ans: **DNA sequences**)
94. **Is Mitosis for growth or reproduction in humans?** (Ans: **Growth and Repair**)

95. **Is Meiosis for growth or reproduction? (Ans: Reproduction/Gives rise to gametes)**
96. **How many daughter cells in Meiosis? (Ans: 4)**
97. **How many daughter cells in Mitosis? (Ans: 2)**
98. **Do prokaryotes have mitochondria? (Ans: No)**
99. **Structural unit of a tissue? (Ans: Cell)**
100. **The ultimate source of energy for cell activity? (Ans: The Sun - through photosynthesis/food)**